

Simulation Model of The Haunted Mansion

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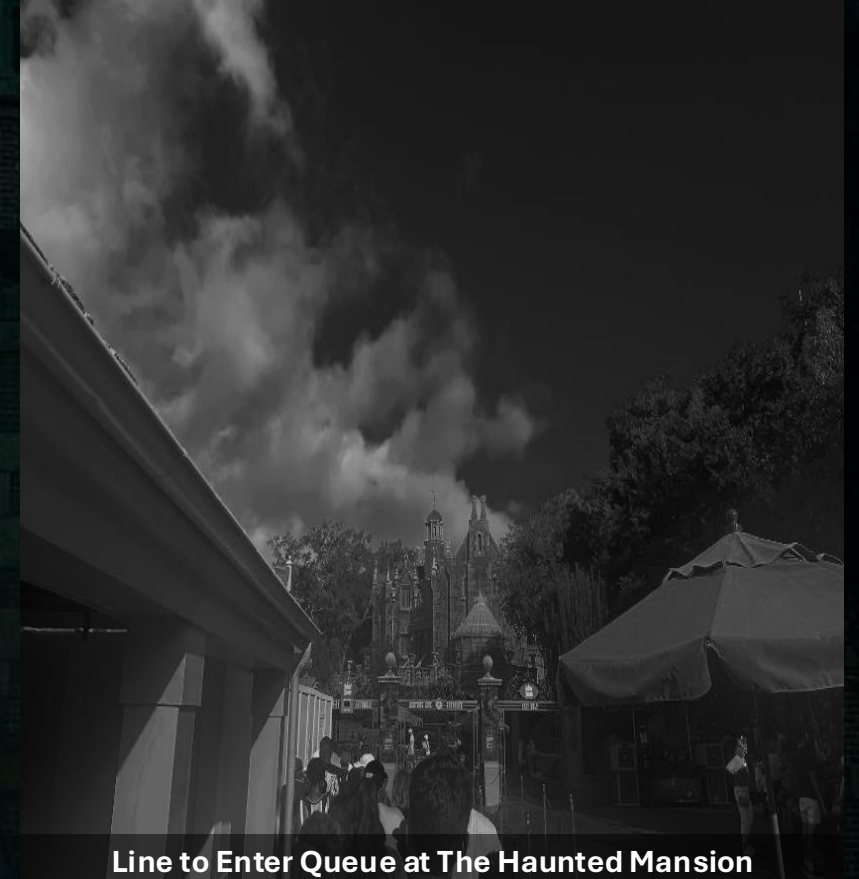
IDS 6145 Simulation Techniques

Dr. Vishnu Prabhu



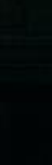
Problem Definition

- Modern amusement parks are growing in popularity and attract millions of visitors per year; guests continue to flock to these attractions to enjoy the thrills, narrative and engaging nature of the rides.
- With a steadily increasing number of visitors year to year, wait times for individual attractions have proportionally increased, leading to longer queue times for rides.
- Our aim in the creation of this project is to assist in resolving the issue of long wait times by optimizing the flow of passengers throughout the duration of an individual attraction.



System Overview

- Our system is a high-fidelity simulation prototype of Walt Disney World's "The Haunted Mansion", complete with adjustable parameters which provides an accurate representation of the entire ride from line entry through exit.
- The system consists of a queue divided into standard and fast pass lanes with an accurate delay for ride duration and maintenance delays, followed by a ride exit.
- This is a standalone system designed to focus on the dynamics and processes of a single attraction; it does not account for human behavior variability or variables which may arise outside the ride such as personnel shortages, weather related park closures or the opening of new rides which may be more popular.



Model Architecture

- Our high-fidelity model begins with a source, comprised of a total source pool of approximately 40,320 daily ride visitors based on a standard 14-hour operating day at Magic Kingdom.
- A select output condition initially divides the source of visitors between lightning pass and non- lightning pass users, starting with a ratio of 50-50 but this can be fine-tuned and changed for further experimentation.
- Both lanes are then merged into the rideGate object, which represents the end of the queue which occurs inside the stretch room preshow.
- Once visitors are inside, several ride delays begin which accounts for the stretch room show, buggy loading and the ride duration.
- Finally, a sink is employed to allow visitors to exit the ride, freeing up more doom buggies for additional guests; the experiment ends once it is cancelled or a threshold of 50,000 guests has been reached.



Doom Buggy Conveyor Belt Loading System

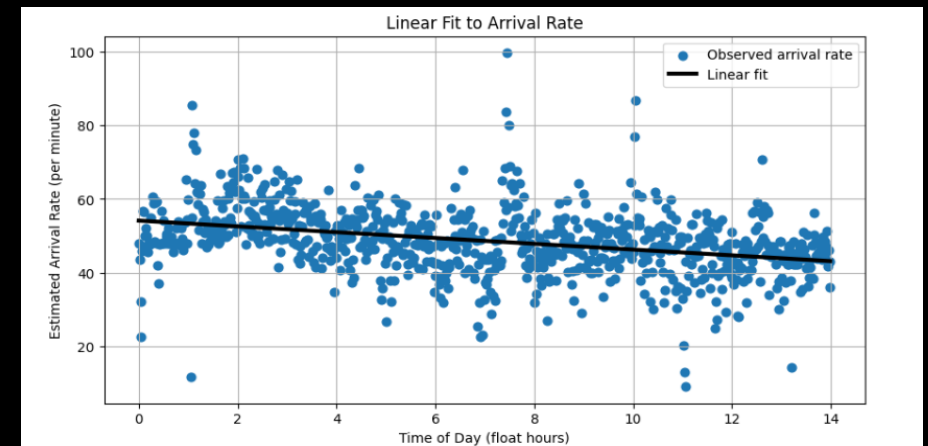
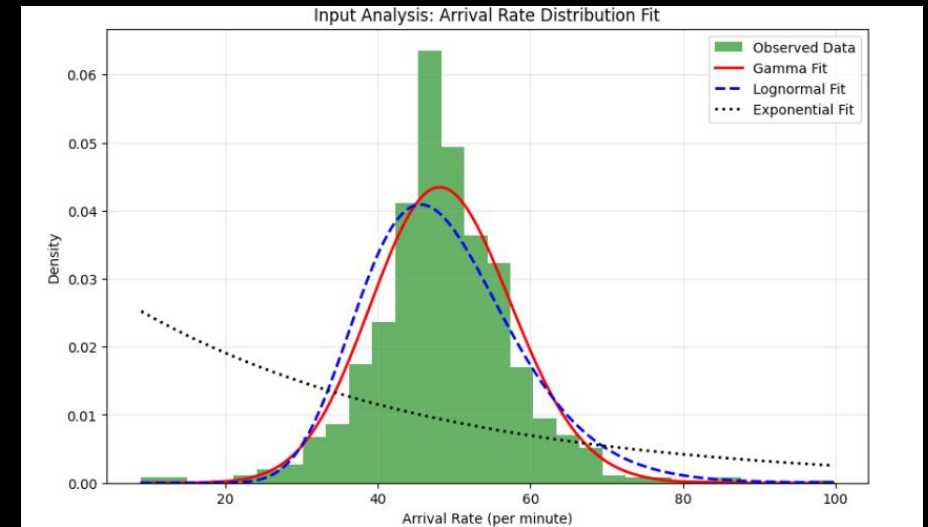


Annotated AnyLogic Model



Input Analysis

- We conducted an input analysis using a Kolmogorov–Smirnov (KS) goodness of fit test, and a Chi-Square goodness of fit test and found the dataset to be an accurate fit for our experiment.
- In general, the dataset follows a predictable and stable pattern of arrival times rather than volatile arrival peaks, indicating that our arrival times are well validated for the model.
- Our model utilizes a time-dependent arrival rate, with a negative slope meaning that over time, λ (arrival intensity of riders) gradually decreases as the day goes on.
- This holds to be accurate because in general, park visitors are much more likely to arrive as the gates open and the weather is cooler rather than later in the day.



Slope: -0.7879026365678381

Intercept: 54.124770696667134



Input Analysis

Failure Duration

- Most failures are short (under 20 minutes) and long outages are rare and the pattern is right-skewed.

Time Between Failures (~1400 mins)

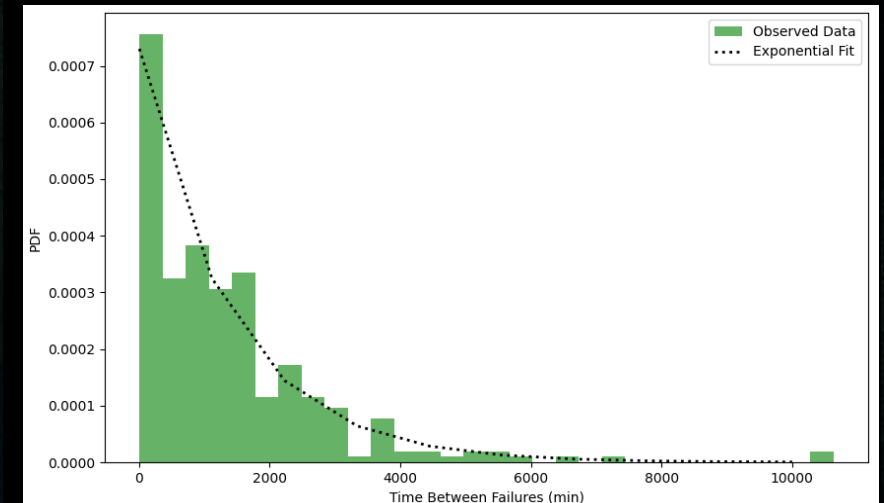
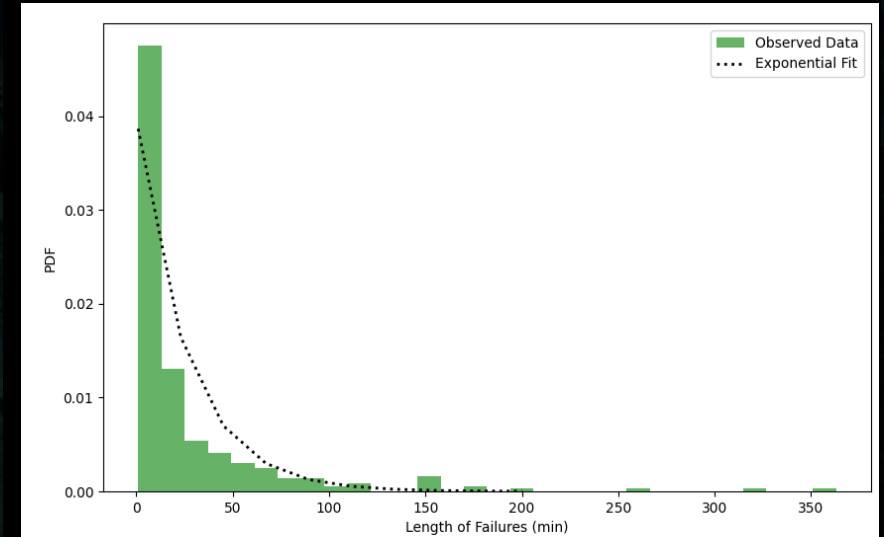
- Historical failure data shows that if a failure happens, it is most likely to occur early, with short downtime.

Distribution Fit

- Exponential distribution closely follows observed data.
- Matches random, memoryless nature of failures.

Model Impact

- Realistic queue buildup during downtimes.
- Better estimation of ride availability and throughput loss.



Verification & Validation

- Verification and validation of the model occurred with a statistically focused visit to the Haunted Mansion, where we found that our actual queue wait time, ride time and number of breakdowns were consistent with our model's findings.
- In addition, our model's operational ride capacity was found to be 2,781 visitors per hour, which closely mirrored that of the OHRC found in Wolfe's estimate of 2880 (Wolfe's maximum capacity is 3200).
- In addition, according to the DarkRide Database, a reliable source for online theme park datasets, the Haunted Mansion's maximum throughput is 3,200 guests per hour.
- We attribute the difference between the maximum (3200) and our operational capacity (2,781) to be wheelchair loading delays and breakdowns. We have a high level of confidence in the model's accuracy based on our heuristics, visits to the ride and independent online databases.

Haunted Mansion Ride Observation

Time to complete ride: 8:55

Stops:

1. 0:30
2. 0:40

Time stopped 70

Total time: 535

Advertised Wait

Time: 13 Mins

Time Entered: 0930 AM

Entrance to Stretch Room: 14 Mins

Time in Stretch: 2 Mins

Stretch Room Ending to Car: 1:38 Mins

Total Stops: 2

Stop #1:

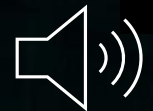
48 Seconds into the Ride, Lasted 30 Seconds

Stop #2

6 Minutes into the Ride, Lasted 43 Seconds

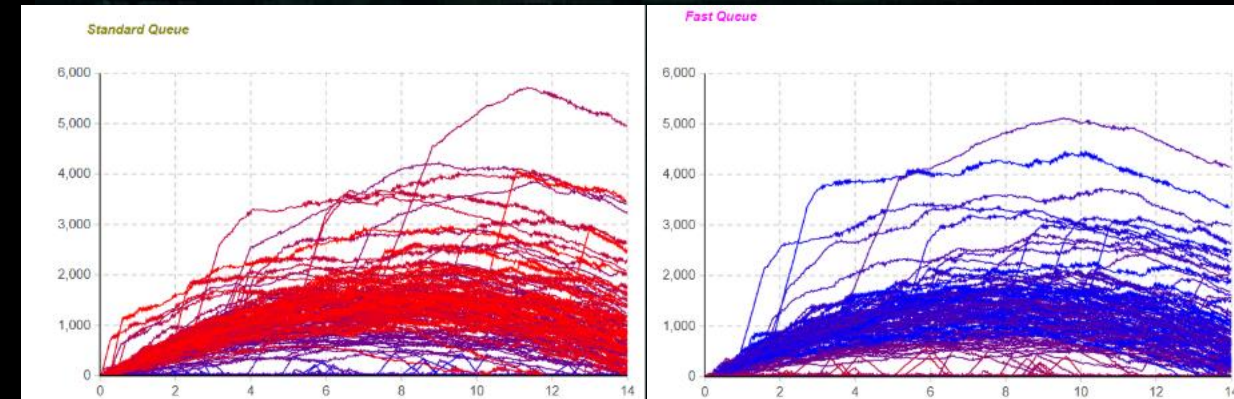
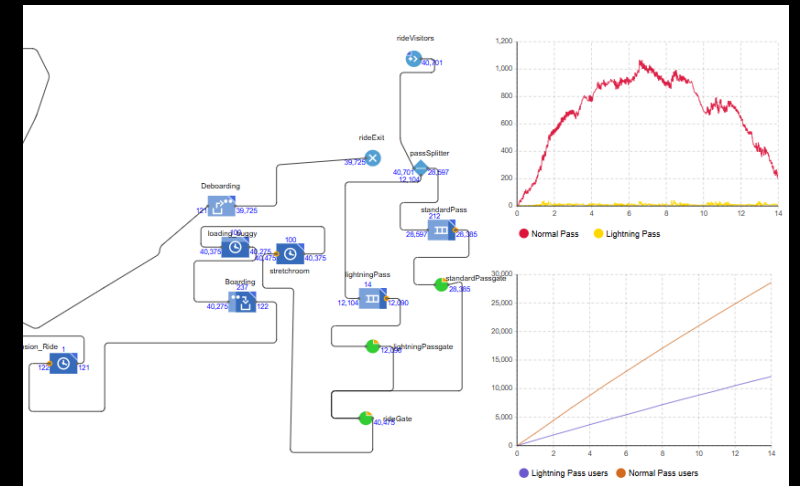
Total Ride Time: 8:5

Queue Photo and Observation 21 October 2025



Results & Analysis

- Our model is a realistic portrayal of the Haunted Mansion's Day to day operational activities; it easily meets demand later in the day, but risks early overload issues in the morning.
- These reliability issues are the largest factor in reaching our 2880 throughput target for each replication.
- After 200 Simulation runs, our model indicates that the probability of guests choosing the Normal queue (pNorm) affects queue distribution but does not significantly change total throughput or ride cycle behavior.
- This confirms that the Haunted Mansion is a capacity limited system: variations in guest behavior redistribute waiting time between Normal and FastPass lines but do not change how many riders are processed.
- Therefore, improving the queue mix does not increase throughput; it only reallocates where delays occur.

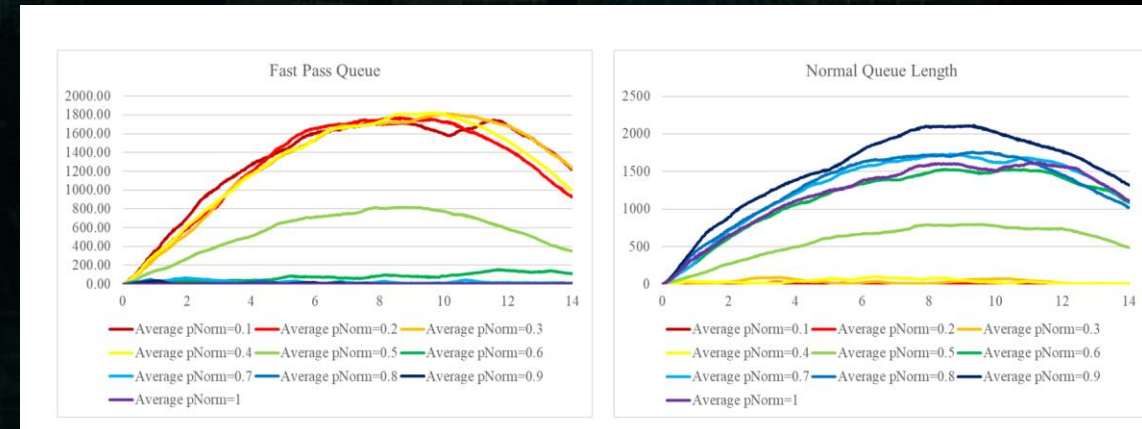
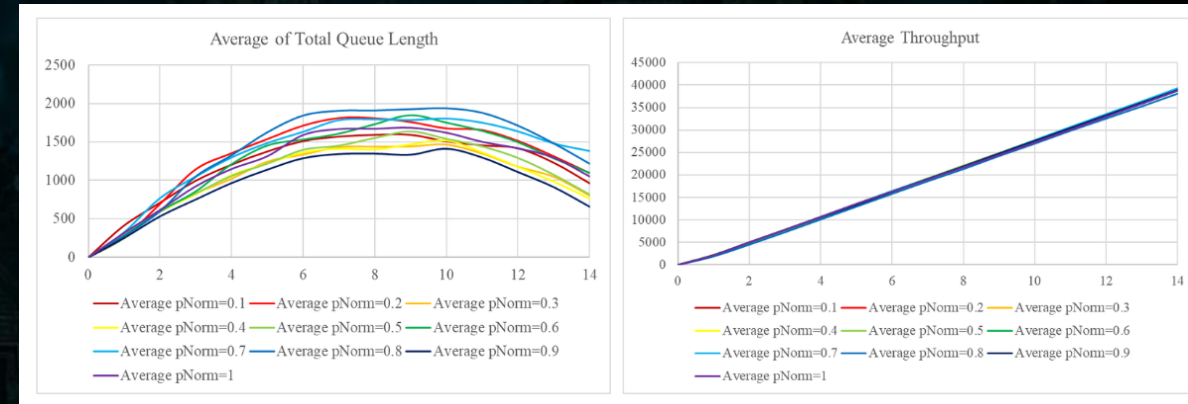


Sample Results After Simulation Run



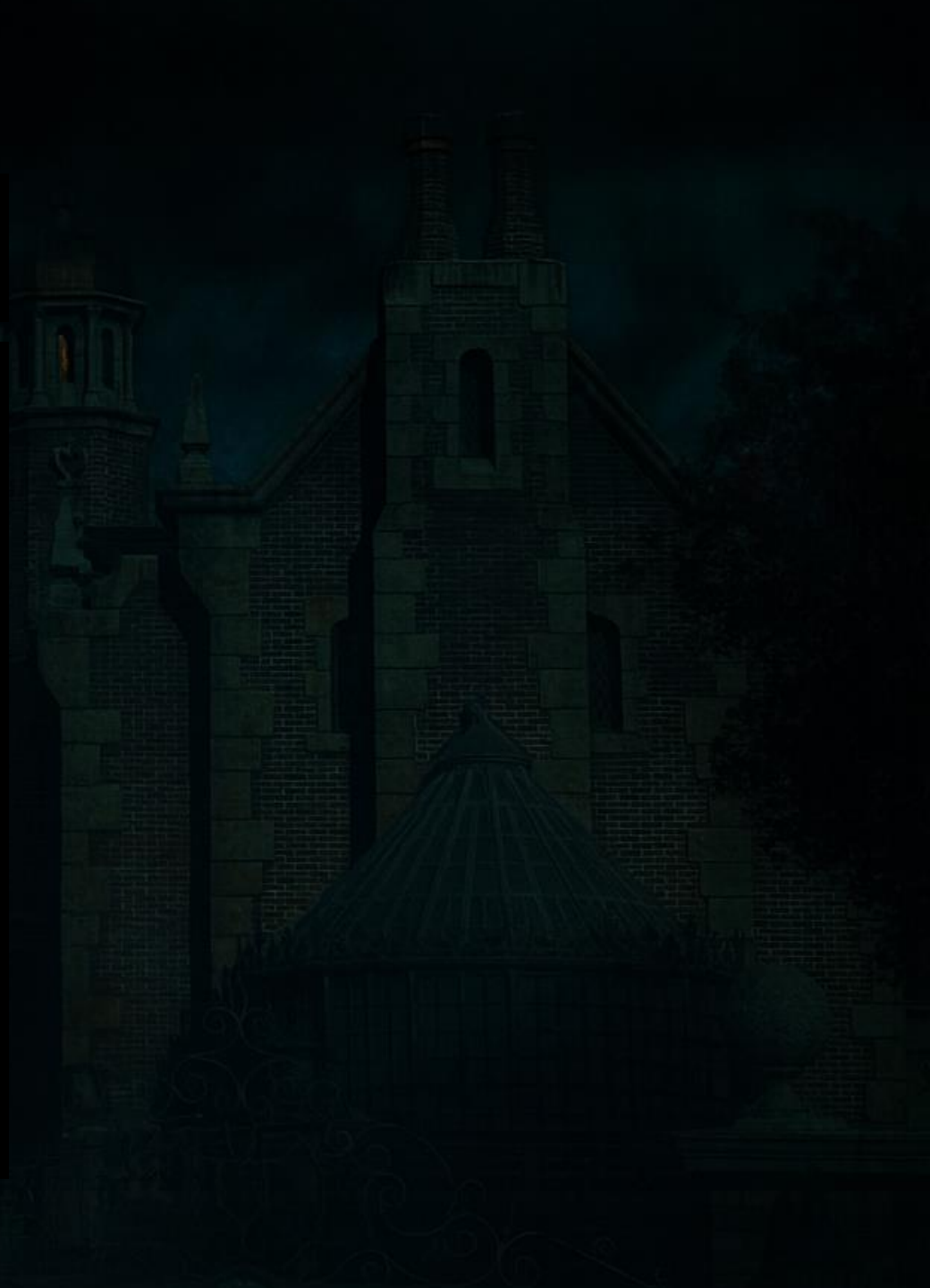
Results & Analysis

- Significant fluctuations were observed across replications due to variations in arrival rates and random ride failures.
- To reduce noise, averages were computed across all replications for each parameter value.
- AnyLogic had difficulties averaging time series data, so data were exported to Excel for post-processing and plotting.
- Averaging reduced noise, making underlying behaviors more apparent.
- Key observed trends:
 - Increasing probability of visitors entering the normal queue → increase in normal queue length.
 - Corresponding decrease in Lightning Pass queue length.
 - Throughput remains largely unchanged.
- Averaged total queue length shows some variation but no clear trend with changes in visitor queue probability.



Conclusions

- This model can be used and implemented by amusement parks and similar venues for visualizing an accurate representation of attraction capabilities and maximum capacity.
- Although large scale parks likely already have systems in place to analyze their rides, our recommendation is that this system be employed by other parks smaller in scale, but with similar attraction objectives. Examples include county fairs, local state-run amusement parks, and circus attractions.
- In addition, we recommend that all parks experiencing longer queue times implement a fast pass system to divide their queues
- Finally, this model can easily be applied to a wide assortment of rides; although it is specific in nature it is easily tailorable to a wide range of attractions.



Future Work

- The system could be improved by implementing a more accurate measurement for the size of the queue in terms of distance. Often during peak hours, attraction workers will open overflow areas to accommodate high volumes of visitors; this logic is not present in our model.
- In addition, the interactions between other rides could also be modeled for a higher level of accuracy in regard to available maintenance personnel; a serious issue at an adjacent ride could remove maintenance personnel from the ride.
- Future research in this area should examine the behavior of guests, adding agent-based modeling. Including these patterns of human behavior would be able to add factors such as satisfaction, dissatisfaction and likelihood to return, and the popularity levels of other attractions. These variables would contribute to an increased level of realism and accuracy for the model.

